

D is false. It should be the change (increase) of internal energy, not the internal energy itself.

- 12 C** $\frac{1}{2} m c^2 = \frac{3}{2} kT$ where T is thermodynamic temperature, c is the r.m.s. speed.

$$\frac{T_f}{T_i} = \frac{c_f^2}{c_i^2}$$

$$\frac{T_f}{127 + 273_i} = \frac{(2c_i)^2}{c_i^2}$$

$$T_f = 1600\text{K}$$

$$\Delta T = 1600 - 400 = 1200\text{K or } 1200^\circ\text{C}$$

- 13 C** Since the graph shows that U increases as the body moves away from the equilibrium position, by